

3(a)	$(\Delta)E = mg(\Delta)h$ or $W(\Delta)h$	C1
	$= 330 \times (4.0 - 1.1)$	A1
	$= 960 \text{ J}$	
3(b)	(work =) $960 - 540$	C1
	(= 420 J)	
	distance moved = $(960 - 540) / 52$ $= 8.1 \text{ m}$	A1
3(c)(i)	$E = \frac{1}{2}mv^2$	C1
	$540 = \frac{1}{2} \times (330 / 9.81) \times v^2$	A1
	$v = 5.7 \text{ m s}^{-1}$	
3(c)(ii)	speed = horizontal component of velocity	C1
	$= 5.7 \times \cos 41^\circ$	
	$= 4.3 \text{ m s}^{-1}$	A1